

# Dr. Casey William Dunn — Curriculum Vitae

## 1 Position and Contact Information

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Professor, Department of Ecology and Evolutionary Biology  
Curator, Division of Invertebrate Zoology, Yale Peabody Museum  
Curator, Department of Informatics, Yale Peabody Museum  
Professor (Secondary), Department of Genetics  
Member, Wu Tsai Institute  
Member, Quantitative Biology Institute (QBio)

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## 2 Education

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| 2005 | Ph.D. Yale University, Department of Ecology and Evolutionary Biology (With Distinction) |
| 2003 | M.S. Yale University, Department of Ecology and Evolutionary Biology                     |
| 1999 | B.S. Stanford University, Biological Sciences (Honors, Phi Beta Kappa)                   |

## 3 Professional Appointments

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| 2026–present | Member, Wu Tsai Institute                                                       |
| 2025–present | Member, Quantitative Biology Institute (QBio)                                   |
| 2021–present | Curator, Informatics, Peabody Museum, Yale University                           |
| 2017–present | Professor, Ecology and Evolutionary Biology, Yale University                    |
| 2019–present | Professor (Secondary), Genetics, Yale University                                |
| 2017–present | Curator, Division of Invertebrate Zoology, Peabody Museum, Yale University      |
| 2017–2018    | Adjunct Professor, Ecology and Evolutionary Biology, Brown University           |
| 2014–2017    | Associate Professor, Ecology and Evolutionary Biology, Brown University         |
| 2010–2014    | Manning Assistant Professor, Ecology and Evolutionary Biology, Brown University |
| 2008–2010    | Assistant Professor, Ecology and Evolutionary Biology, Brown University         |

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|-----------|-----------------------------------------------------------------------------------------------|
| 2007–2008 | Assistant Professor (Research), Ecology and Evolutionary Biology, Brown University            |
| 2005–2007 | Postdoctoral Fellow, Kewalo Marine Lab, University of Hawaii                                  |
| 1999–2000 | Research Assistant with Stuart Chapin III and James Randerson, University of Alaska Fairbanks |
| 1995–1998 | Co-op Technical at KLA-Tencor.                                                                |
| 1994–1995 | Tour guide at La Selva Jungle Lodge in the Amazon Basin of eastern Ecuador                    |
| 1993–1994 | Student researcher with John Postlethwait, University of Oregon                               |

## 4 Publications

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### a. Books

- Dunn, C. W. (2026). *Phylogenetic Biology*. First Edition. New Haven, CT: Casey W. Dunn. ISBN: 979-8-9934524-0-1. DOI: 10.5281/zenodo.17267993. URL: [https://dunnlab.org/phylogenetic\\_biology/](https://dunnlab.org/phylogenetic_biology/).
- Haddock, S. H. D. and C. W. Dunn (2010). *Practical Computing for Biologists*. Sunderland, MA: Sinauer Associates, p. 538. URL: <http://practicalcomputing.org>.

### b. Chapters in books

- Collins, A. G., M. Daly, and C. W. Dunn (2020). “Cnidaria”. In: *Phylonyms: A Companion to the PhyloCode*. CRC Press, pp. 457–460. DOI: 10.1201/9780429446276.
- Collins, A. G. and C. W. Dunn (2020). “Medusozoa”. In: *Phylonyms: A Companion to the PhyloCode*. CRC Press, pp. 473–475. DOI: 10.1201/9780429446276.
- Dunn, C. W. (2020). “Siphonophora”. In: *Phylonyms: A Companion to the PhyloCode*. CRC Press, pp. 491–492. DOI: 10.1201/9780429446276.
- Dunn, C. W. and A. G. Collins (2020a). “Hydroidolina”. In: *Phylonyms: A Companion to the PhyloCode*. CRC Press, pp. 489–490. DOI: 10.1201/9780429446276.
- Dunn, C. W. and A. G. Collins (2020b). “Hydrozoa”. In: *Phylonyms: A Companion to the PhyloCode*. CRC Press, pp. 473–475. DOI: 10.1201/9780429446276.
- Dunn, C. W. and A. G. Collins (2020c). “Trachylina”. In: *Phylonyms: A Companion to the PhyloCode*. CRC Press, pp. 485–487. DOI: 10.1201/9780429446276.
- Giribet, G., C. W. Dunn, G. D. Edgecombe, A. Hejnol, M. Q. Martindale, and G. W. Rouse (2009). “Assembling the spiralian tree of life”. In: *Animal Evolution*. Oxford University Press, pp. 52–64.
- Mills, C. E., A. C. Marques, A. E. Migotto, D. R. Calder, C. Hand, J. T. Rees, S. H. D. Haddock, C. W. Dunn, and P. R. Pugh (2007). “Hydrozoa: Polyps, Hydromedusae, and Siphonophora”. In: *The Light & Smith manual: intertidal invertebrates from central California to Oregon, 4th edition*. University of California Press.

### c. Refereed journal articles

- Ahuja, N., D. T. Schultz, D. Destanović, S. H. Church, N. Picciani, C. Munro, K. Kon-Nanjo, T. Kon, M. K. Mañko, W. Shi, O. Simakov, and C. W. Dunn (July 2026). “Siphonophore genome structure and the evolution of functional specialization”. In: *PLOS ONE* 21.7, pp. 1–25. DOI: 10.1371/journal.pone.0351247.
- Dunn, C. W. and S. H. Church (Apr. 2026). “Sharkmer: repurposing PCR primers for targeted genome assembly using in silico PCR”. In: *Bioinformatics*, btag163. ISSN: 1367-4811. DOI: 10.1093/bioinformatics/btag163.

- Church, S. H., R. B. Abedon, N. Ahuja, C. J. Anthony, D. Destanović, D. A. Ramirez, L. M. Rojas, M. E. Albinsson, I. Álvarez Trasobares, R. E. Bergemann, O. Bogdanovic, D. R. Burdick, T. J. Cunha, A. Damian-Serrano, G. D'Elia, K. B. Dion, T. K. Doyle, J. M. Gonçalves, A. Gonzalez Rajal, S. H. D. Haddock, R. R. Helm, D. Le Gouvello, Z. R. Lewis, B. I. M. M. Magalhães, M. K. Mañko, C. G. Mayorga-Adame, A. de Mendoza, C. J. Moura, C. Munro, R. Nel, K. Oguchi, J. N. Perelman, L. Prieto, K. A. Pitt, M. Roughan, A. Schaeffer, A. L. Schmidt, J. Sellanes, N. G. Wilson, G. Yamamoto, E. A. Lazo-Wasem, C. Simon, M. B. Decker, J. M. Coughlan, and C. W. Dunn (2025). "Population genomics of a sailing siphonophore reveals genetic structure in the open ocean". In: *Current Biology* 35 (15), 3556–3569.e6. doi: 10.1016/j.cub.2025.05.066.
- Guang, A. A., C. W. Dunn, V. Novitsky, M. Howison, and R. Kantor (2025). "T-shaped alignments integrating HIV-1 near full-length genome and partial pol sequences can improve phylogenetic inference of transmission clusters". In: *PLOS Computational Biology* 21.11, pp. 1–18. doi: 10.1371/journal.pcbi.1013676.
- Oguchi, K., A. Maeno, K. Yoshida, H. Kohtsuka, and C. W. Dunn (2025). "Zooid arrangement and colony growth in *Porpita porpita*". In: *Frontiers in Zoology* 22, p. 11. doi: 10.1186/s12983-025-00565-3.
- Price, K. L., D. M. Tharakan, W. Salvenmoser, K. Ayers, J. Mah, C. W. Dunn, B. Hobmayer, and L. Cooley (2025). "Examination of Germline and Somatic Intercellular Bridges in *Hydra vulgaris* Reveals Insights into the Evolutionarily Conserved Mechanism of Intercellular Bridge Formation". In: *Molecular Biology and Evolution* 42, msaf248. doi: 10.1093/molbev/msaf248.
- Sebé-Pedrós, A., A. Tanay, M. K. N. Lawniczak, D. Arendt, S. Aerts, J. Archibald, M. I. Arnone, M. Blaxter, P. Cleves, S. M. Coelho, M. Dias, C. W. Dunn, A. Elek, J. Frazer, T. Gabaldón, J. Gillis, X. Grau-Bové, R. Guigó, O. Hobert, J. Huerta-Cepas, M. Irimia, A. Klein, H. Lewin, C. J. Lowe, H. Marlow, J. M. Musser, L. G. Nagy, S. R. Najle, L. Pachter, S. Paez, I. Papatheodorou, M. J. Passalacqua, N. Rajewsky, S. Y. Rhee, T. A. Richards, T. Sauka-Spengler, L. M. Saunders, E. Seuntjens, J. Solana, Y. Song, U. Technau, and B. Wang (2025). "The Biodiversity Cell Atlas: mapping the tree of life at cellular resolution". In: *Nature* 645, pp. 877–885. doi: 10.1038/s41586-025-09312-4.
- Ahuja, N., X. Cao, D. T. Schultz, N. Picciani, A. Lord, S. Shao, K. Jia, D. R. Burdick, S. H. D. Haddock, Y. Li, and C. W. Dunn (2024). "Giants among Cnidaria: Large Nuclear Genomes and Rearranged Mitochondrial Genomes in Siphonophores". In: *Genome Biology and Evolution* 16 (3). doi: 10.1093/gbe/evae048.
- Church, S. H., J. L. Mah, and C. W. Dunn (2024). "Integrating phylogenies into single-cell RNA sequencing analysis allows comparisons across species, genes, and cells". In: *PLOS Biology* 22, e3002633. doi: 10.1371/journal.pbio.3002633.
- Fulton, J., V. Novitsky, F. Gillani, A. Guang, J. Steingrímsson, A. Khanna, J. Hague, C. Dunn, J. Hogan, K. Howe, M. MacAskill, L. Bhattarai, T. Bertrand, U. Bandy, and R. Kantor (2024). "Integrating HIV Cluster Analysis in Everyday Public Health Practice: Lessons Learned From a Public Health–Academic Partnership". In: *JAIDS Journal of Acquired Immune Deficiency Syndromes* 97, pp. 48–54. doi: 10.1097/QAI.0000000000003469.
- Hetherington, E. D., H. G. Close, S. H. Haddock, A. D. Damian-Serrano, C. W. Dunn, N. J. Wallsgrave, S. C. Doherty, and C. A. Choy (2024). "Vertical trophic structure and niche partitioning of gelatinous predators in a pelagic food web: Insights from stable isotopes of siphonophores". In: *Limnology and Oceanography* 69 (4), pp. 902–919. doi: 10.1002/lno.12536.
- Kantor, R., J. Steingrímsson, J. Fulton, V. Novitsky, M. Howison, F. Gillani, L. Bhattarai, M. MacAskill, J. Hague, A. Guang, A. Khanna, C. Dunn, J. Hogan, T. Bertrand, and U. Bandy (2024). "Prospective evaluation of routine statewide integration of molecular epidemiology and contact tracing to disrupt HIV transmission". In: *Open Forum Infectious Diseases* 10 (11). doi: 10.1093/ofid/ofae599.
- Mah, J. L. and C. W. Dunn (2024). "Cell type evolution reconstruction across species through cell phylogenies of single-cell RNA sequencing data". In: *Nature Ecology and Evolution* 8, pp. 325–338. doi: 10.1038/s41559-023-02281-9.
- Oguchi, K., G. Yamamoto, H. Kohtsuka, and C. W. Dunn (2024). "Physalia gonodendra are not yet sexually mature when released". In: *Scientific Reports* 14, p. 23011. doi: 10.1038/s41598-024-73611-5.
- Church, S. H., J. L. May, G. Wagner, and C. W. Dunn (2023). "Normalizing need not be the norm: count-based math for analyzing single-cell data". In: *Theory in Biosciences*. doi: 10.1007/s12064-023-00408-x.
- Church, S. H., C. Munro, C. W. Dunn, and C. G. Extavour (2023). "The evolution of ovary-biased gene expression in Hawaiian *Drosophila*". In: *PLoS Genet* 19, e1010607. doi: 10.1371/journal.pgen.1010607.
- Dunn, C. W. (2023). "Neurons that connect without synapses". In: *Science* 380, pp. 241–242. doi: 10.1126/science.adh0542.

- Howison, M., F. S. Gillani, V. Novitsky, J. A. Steingrimsson, J. Fulton, T. Bertrand, K. Howe, A. Civitarese, L. Bhattarai, M. MacAskill, G. Ronquillo, J. Hague, C. W. Dunn, U. Bandy, J. W. Hogan, and R. Kantor (2023). “An Automated Bioinformatics Pipeline Informing Near-Real-Time Public Health Responses to New HIV Diagnoses in a Statewide HIV Epidemic”. In: *Viruses* 15, p. 737. URL: <https://www.mdpi.com/1999-4915/15/3/737>.
- Khanna, A., V. Novitsky, A. Guang, M. Howison, F. S. Gillani, J. Steingrimsson, C. Dunn, J. Fulton, T. Bertrand, K. Howe, L. Bhattarai, G. Ronquillo, M. MacAskill, U. Bandy, and R. Kantor (2023). “Integrating HIV Partner Services and Molecular Epidemiology Data to Enhance HIV Transmission Disruption in Rhode Island: Findings from a Public Health-Academic Partnership”. In: *Open Forum Infectious Diseases* 10. doi: 10.1093/ofid/ofad500.170.
- Novitsky, V., J. Steingrimsson, M. Howison, C. W. Dunn, F. S. Gillani, J. Fulton, T. Bertrand, K. Howe, L. Bhattarai, G. Ronquillo, M. MacAskill, U. Bandy, J. Hogan, and R. Kantor (2023). “Not all clusters are equal: dynamics of molecular HIV-1 clusters in a statewide Rhode Island epidemic”. In: *AIDS*. doi: 10.1097/QAD.0000000000003426.
- Damian-Serrano, A., E. D. Hetherington, A. Choy, S. H. D. Haddock, A. Lepides, and C. W. Dunn (2022). “Characterizing the secret diets of siphonophores (Cnidaria: Hydrozoa) using DNA metabarcoding”. In: *PLoS One*. doi: 10.1371/journal.pone.0267761.
- Guang, A., M. Howison, L. Ledingham, M. D’Antuono, P. A. Chan, C. Lawrence, C. W. Dunn, and R. Kantor (2022). “Incorporating Within-Host Diversity in Phylogenetic Analyses for Detecting Clusters of New HIV Diagnoses”. In: *Frontiers in Microbiology*. doi: 10.3389/fmicb.2021.803190.
- Hetherington, E. D., A. Damian-Serrano, S. H. D. Haddock, C. W. Dunn, and A. Choy (2022). “Integrating siphonophores into marine food-web ecology”. In: *Limnology and Oceanography Letters*. doi: 10.1002/lol2.10235.
- Munro, C., F. Zapata, M. Howison, S. Siebert, and C. W. Dunn (2022). “Evolution of gene expression across species and specialized zooids in Siphonophora”. In: *MBE*. doi: 10.1093/molbev/msac027.
- Steingrimsson, J. A., J. Fulton, M. H. Howison, V. Novitsky, F. S. Gillani, T. Bertrand, A. Civitarese, K. Howe, G. Ronquillo, B. Lafazia, Z. Parillo, T. Marak, P. A. Chan, L. Bhattarai, C. W. Dunn, U. Bandy, N. A. Scott, J. W. Hogan, and R. Kantor (2022). “Beyond HIV outbreaks: protocol, rationale, and implementation of a prospective study quantifying the benefit of incorporating viral sequence clustering analysis into routine public health interventions”. In: *BMJ Open*. doi: 10.1136/bmjopen-2021-060184.
- Damian-Serrano, A., S. H. D. Haddock, and C. W. Dunn (2021a). “The evolution of siphonophore tentilla for specialized prey capture in the open ocean”. In: *PNAS* 118, e2005063118. doi: 10.1073/pnas.2005063118.
- Damian-Serrano, A., S. H. D. Haddock, and C. W. Dunn (2021b). “The evolutionary history of siphonophore tentilla: Novelties, convergence, and integration”. In: *Integrative Organismal Biology*. doi: 10.1093/iob/obab019.
- Guang, A., M. Howison, F. Zapata, C. Lawrence, and C. W. Dunn (2021). “Revising transcriptome assemblies with phylogenetic information”. In: *PLoS ONE* 16, e0244202. doi: 10.1371/journal.pone.0244202.
- Li, Y., X. X. Shen, B. Evans, C. W. Dunn, and A. Rokas (2021). “Rooting the animal tree of life”. In: *MBE*. doi: 10.1093/molbev/msab170.
- Li, Y., J. L. Steenwyk, Y. Chang, Y. Wang, T. Y. James, J. E. Stajich, J. W. Spatafora, M. Groenewald, C. W. Dunn, C. T. Hittinger, X. X. Shen, and A. Rokas (2021). “A genome-scale phylogeny of the kingdom Fungi”. In: *Current Biology* 31, pp. 1653–1665. doi: 10.1016/j.cub.2021.01.074.
- Novitsky, V., J. Steingrimsson, M. Howison, C. W. Dunn, F. S. Gillani, A. Manne, Y. Li, M. Spence, Z. Parillo, J. Fulton, T. Marak, P. Chan, T. Bertrand, U. Bandy, N. Alexander-Scott, J. Hogan, and R. Kantor (2021). “Longitudinal typing of molecular HIV clusters in a statewide epidemic”. In: *AIDS*. doi: 10.1097/QAD.0000000000002953.
- Kantor, R., J. P. Fulton, J. Steingrimsson, V. Novitsky, M. Howison, F. Gillani, Y. Li, A. Manne, Z. Parillo, M. Spence, T. Marak, P. Chan, C. W. Dunn, T. Bertrand, U. Bandy, N. Alexander-Scott, and J. W. Hogan (2020). “Challenges in evaluating the use of viral sequence data to identify HIV transmission networks for public health”. In: *Statistical Communications in Infectious Diseases* 12, s1. doi: 10.1515/scid-2019-0019.
- Novitsky, V., J. A. Steingrimsson, M. Howison, F. S. Gillani, Y. Li, A. Manne, J. Fulton, M. Spence, Z. Parillo, T. Marak, P. A. Chan, T. Bertrand, U. Bandy, N. Alexander-Scott, C. W. Dunn, J. Hogan, and R. Kantor (2020). “Empirical comparison of analytical approaches for identifying molecular HIV-1 clusters”. In: *Scientific Reports* 10. doi: 10.1038/s41598-020-75560-1.
- Smith, S. D., M. W. Pennell, C. W. Dunn, and S. V. Edwards (2020). “Phylogenetics is the New Genetics (for Most of Biodiversity)”. In: *Trends in Ecology and Evolution*. doi: 10.1016/j.tree.2020.01.005.

- Munro, C., Z. Vue, R. R. Behringer, and C. W. Dunn (2019). “Morphology and development of the Portuguese man of war, *Physalia physalis*”. In: *Scientific Reports* 9, p. 15522. doi: 10.1038/s41598-019-51842-1.
- Wang, L. G., T. T. Y. Lam, S. Xu, Z. Dai, L. Zhou, T. Feng, P. Guo, C. W. Dunn, B. R. Jones, T. Bradley, H. Zhu, Y. Guan, Y. Jiang, and G. Yu (2019). “treeio: an R package for phylogenetic tree input and output with richly annotated and associated data”. In: *Molecular Biology and Evolution*. doi: 10.1093/molbev/msz240.
- Dunn, C. W., F. Zapata, C. Munro, S. Siebert, and A. Hejnol (2018). “Pairwise comparisons across species are problematic when analyzing functional genomic data”. In: *PNAS*. doi: 10.1073/pnas.1707515115.
- Haddock, S. H. D., L. M. Christianson, W. R. Francis, S. Martini, C. W. Dunn, P. R. Pugh, C. E. Mills, K. J. Osborn, B. A. Seibel, C. A. Choy, C. E. Schnitzler, G. I. Matsumoto, M. Messi, D. T. Schultz, J. R. Winnikoff, M. L. Powers, R. Gasca, W. E. Browne, S. Johnsen, K. L. Schlining, S. von Thun, B. E. Erwin, J. F. Ryan, and E. V. Thuesen (2018). “Insights into the Biodiversity, Behavior, and Bioluminescence of Deep-Sea Organisms Using Molecular and Maritime Technology”. In: *Oceanography* 30, pp. 38–47. doi: 10.5670/oceanog.2017.422.
- Munro, C., S. Siebert, F. Zapata, M. Howison, A. Damian-Serrano, S. H. Church, F. E. Goetz, P. R. Pugh, S. H. D. Haddock, and C. W. Dunn (2018). “Improved phylogenetic resolution within Siphonophora (Cnidaria) with implications for trait evolution”. In: *Molecular Phylogenetics and Evolution*. doi: 10.1016/j.ympev.2018.06.030.
- Pugh, P. R., C. W. Dunn, and S. H. D. Haddock (2018). “Description of *Tottonophyes enigmatica* gen. nov., sp. nov. (Hydrozoa, Siphonophora, Calycophorae), with a reappraisal of the function and homology of nectophoral canals”. In: *Zootaxa* 4415, pp. 452–472. doi: 10.11646/zootaxa.4415.3.3.
- Helm, R. R. and C. W. Dunn (2017). “Indoles induce metamorphosis in a broad diversity of jellyfish, but not in a crown jelly (Coronatae)”. In: *PLoS One*. doi: 10.1371/journal.pone.0188601.
- Dunn, C. W. and C. Munro (2016). “Comparative genomics and the diversity of life”. In: *Zoologica Scripta* 45, pp. 5–13. doi: 10.1111/zsc.12211.
- Guang, A., F. Zapata, M. Howison, C. E. Lawrence, and C. W. Dunn (2016). “An Integrated Perspective on Phylogenetic Workflows”. In: *Trends in Ecology and Evolution* 31, pp. 116–126. doi: 10.1016/j.tree.2015.12.007.
- Church, S. H., J. F. Ryan, and C. W. Dunn (2015). “Automation and Evaluation of the SOWH Test of Phylogenetic Topologies with SOWHAT”. In: *Systematic Biology* 64, pp. 1048–1058. doi: 10.1093/sysbio/syv055.
- Church, S. H., S. Siebert, P. Bhattacharyya, and C. W. Dunn (2015). “The Histology of *Nanomia bijuga* (Hydrozoa: Siphonophora)”. In: *J. Exp. Zool. (Mol. Dev. Evol.)* 324, pp. 435–449. doi: 10.1002/jez.b.22629.
- Dunn, C. W. (2015). “Acorn worms in a nutshell”. In: *Nature* 527, pp. 448–449. doi: 10.1038/nature16315.
- Dunn, C. W., S. P. Leys, and S. H. D. Haddock (2015). “The hidden biology of sponges and ctenophores”. In: *Trends in Ecology and Evolution* 30, pp. 282–291. doi: 10.1016/j.tree.2015.03.003.
- Dunn, C. W. and J. T. Ryan (2015). “The evolution of animal genomes”. In: *Current Opinion in Genetics and Development* 35, pp. 25–32. doi: 10.1016/j.gde.2015.08.006.
- Gonzalez, V. L., S. C. S. Andrade, R. Bieler, T. M. Collins, C. W. Dunn, P. M. Mikkelsen, J. D. Taylor, and G. Giribet (2015). “A phylogenetic backbone for Bivalvia: an RNA-seq approach”. In: *Proceedings of the Royal Society B: Biological Sciences* 282, p. 20142332. doi: 10.1098/rspb.2014.2332.
- Haddock, S. H. and C. W. Dunn (2015). “Fluorescent proteins function as a prey attractant: experimental evidence from the hydromedusa *Olindias formosus* and other marine organisms”. In: *Biology Open* 4, pp. 1094–1104. doi: 10.1242/bio.012138.
- Helm, R. R., S. Tiozzo, M. K. S. Lilley, F. Lombard, and C. W. Dunn (2015). “Comparative muscle development of scyphozoan jellyfish with simple and complex life cycles”. In: *EvoDevo* 6, p. 11. doi: 10.1186/s13227-015-0005-7.
- Laumer, C. E., N. Bekkouche, A. Kerbl, F. Goetz, R. C. Neves, M. V. Sorensen, R. M. Kristensen, A. Hejnol, C. W. Dunn, G. Giribet, and K. Worsaae (2015). “Spiralian Phylogeny Informs the Evolution of Microscopic Lineages”. In: *Current Biology* 25, pp. 2000–2006. doi: 10.1016/j.cub.2015.06.068.
- Reich, A., C. W. Dunn, K. Akasaka, and G. Wessel (2015). “Phylogenomic Analyses of Echinodermata Support the Sister Groups of Asterozoa and Echinozoa”. In: *PLoS One* 10, e0119627. doi: 10.1371/journal.pone.0119627.
- Salinas-Saavedra, M., T. Q. Stephenson, C. W. Dunn, and M. Q. Martindale (2015). “Par system components are asymmetrically localized in ectodermal epithelia, but not during early development in the sea anemone *Nematostella vectensis*”. In: *EvoDevo* 6, p. 20. doi: 10.1186/s13227-015-0014-6.
- Siebert, S., F. E. Goetz, S. H. Church, P. Bhattacharyya, F. Zapata, S. H. D. Haddock, and C. W. Dunn (2015). “Stem Cells in a Colonial Animal with Localized Growth Zones”. In: *EvoDevo* 6, p. 22. doi: 10.1186/s13227-015-0018-2.

- Zapata, F., F. E. Goetz, S. A. Smith, M. Howison, S. Siebert, S. Church, S. M. Sanders, C. L. Ames, C. S. McFadden, S. C. France, M. Daly, A. G. Collins, S. H. D. Haddock, C. W. Dunn, and P. Cartwright (2015). "Phylogenomic analyses support traditional relationships within Cnidaria". In: *PLoS One* 10, e0139068. doi: 10.1371/journal.pone.0139068.
- Dunn, C. W. (2014g). "Reconsidering the phylogenetic utility of miRNA in animals". In: *PNAS* 111, pp. 12576–12577. doi: 10.1073/pnas.1413545111.
- Dunn, C. W., G. Giribet, G. D. Edgecombe, and A. Hejnol (2014). "Animal Phylogeny and its Evolutionary Implications". In: *Annual Review of Ecology, Evolution, and Systematics* 45, pp. 371–395. doi: 10.1146/annurev-ecolsys-120213-091627.
- Howison, M., F. Zapata, E. J. Edwards, and C. W. Dunn (2014). "Bayesian genome assembly and assessment by Markov Chain Monte Carlo sampling". In: *PLoS One* 9, e99497. doi: 10.1371/journal.pone.0099497.
- Lopez, J., H. Bracken-Grissom, A. Collins, T. Collins, K. Crandall, D. Distel, C. Dunn, and et al. (2014). "Global Invertebrate Genomics Alliance (GIGA): Developing Community Resources to Study Diverse Invertebrate Genomes". In: *Journal of Heredity* 105, pp. 1–18. doi: 10.1093/jhered/est084.
- Zapata, F., N. G. Wilson, M. Howison, S. C. S. Andrade, K. M. Jörger, M. Schrödl, F. E. Goetz, G. Giribet, and C. W. Dunn (2014). "Phylogenomic analyses of deep gastropod relationships reject Orthogastropoda". In: *Proceedings of the Royal Society B: Biological Sciences* 281, pp. 1471–2954. doi: 10.1098/rspb.2014.1739.
- Dunn, C. W., M. Howison, and F. Zapata (2013). "Agalma: an automated phylogenomics workflow". In: *BMC Bioinformatics* 14, p. 330. doi: 10.1186/1471-2105-14-330.
- Dunn, C. W., X. Luo, and Z. Wu (2013). "Phylogenetic analysis of gene expression". In: *Integrative and Comparative Biology* 53, pp. 847–856. doi: 10.1093/icb/ict068.
- Helm, R. R., S. Siebert, S. Tulin, J. Smith, and C. W. Dunn (2013). "Characterization of differential transcript abundance through time during *Nematostella vectensis* development". In: *BMC Genomics* 14, p. 266. doi: 10.1186/1471-2164-14-266.
- Howison, M., F. Zapata, and C. W. Dunn (2013). "Toward a statistically explicit understanding of de novo sequence assembly". In: *Bioinformatics* 29, pp. 2959–2963. doi: 10.1093/bioinformatics/btt525.
- Ryan, J. F., K. Pang, C. E. Schnitzler, A. Nguyen, R. T. Moreland, D. K. Simmons, B. J. Koch, W. R. Francis, P. Havlak, N. I. S. C. Comparative Sequencing Program, S. A. Smith, N. H. Putnam, S. H. D. Haddock, C. W. Dunn, T. G. Wolfsberg, J. C. Mullikin, M. Q. Martindale, and A. D. Baxeavanis (2013). "The genome of the ctenophore *Mnemiopsis leidyi* and its implications for cell type evolution". In: *Science* 432, p. 1242592. doi: 10.1126/science.1242592.
- Siebert, S., P. R. Pugh, S. H. D. Haddock, and C. W. Dunn (2013). "Re-evaluation of characters in Apolemiidae (Siphonophora), with description of two new species from Monterey Bay, California". In: *Zootaxa* 3702, pp. 201–232. URL: <http://www.mapress.com/zootaxa/2013/f/zt03702p232.pdf>.
- Howison, M., N. Sinnott-Armstrong, and C. W. Dunn (2012). "BioLite, a lightweight bioinformatics framework with automated tracking of diagnostics and provenance". In: *4th USENIX Workshop on the Theory and Practice of Provenance (TaPP12)*. URL: <https://www.usenix.org/system/files/conference/tapp12/tapp12-final5.pdf>.
- Edgecombe, G. D., G. Giribet, C. W. Dunn, A. Hejnol, R. M. Kristensen, R. C. Neves, G. W. Rouse, K. Worsaae, and M. V. Sørensen (2011). "Higher-level metazoan relationships: recent progress and remaining questions". In: *Organisms, Diversity, and Evolution* 11, pp. 151–172. doi: 10.1007/s13127-011-0044-4.
- Siebert, S., M. D. Robinson, S. C. Tintori, F. Goetz, R. R. Helm, S. A. Smith, N. Shaner, S. H. D. Haddock, and C. W. Dunn (2011). "Differential Gene Expression in the Siphonophore *Nanomia bijuga* (Cnidaria) Assessed with Multiple Next-Generation Sequencing Workflows". In: *PLoS One* 6, e22953. doi: 10.1371/journal.pone.0022953.
- Smith, S. A., N. G. Wilson, F. Goetz, C. Feehery, S. C. S. Andrade, G. W. Rouse, G. Giribet, and C. W. Dunn (2011). "Resolving the evolutionary relationships of molluscs with phylogenomic tools". In: *Nature* 480, pp. 364–367. doi: 10.1038/nature10526.
- Hejnol, A., M. Obst, A. Stamatakis, M. Ott, G. Rouse, G. Edgecombe, P. Martinez, J. Baguña, X. Bailly, U. Jondelius, M. Wiens, W. E. G. Müller, E. Seaver, W. C. Wheeler, M. Q. Martindale, G. Giribet, and C. W. Dunn (2009). "Assessing the root of bilaterian animals with scalable phylogenomic methods". In: *Proc. R. Soc. B* 276, pp. 4261–4270. doi: 10.1098/rspb.2009.0896.

- Cartwright, P., N. M. Evans, C. W. Dunn, A. C. Marques, M. P. Miglietta, P. Schuchert, and A. G. Collins (2008). "Phylogenetics of Hydroidolina (Hydrozoa: Cnidaria)". In: *Journal of the Marine Biological Association of the United Kingdom* 88, pp. 1663–1672. doi: 10.1017/S0025315408002257.
- Dunn, C. W., A. Hejnol, D. Q. Matus, K. Pang, W. E. Browne, S. A. Smith, E. Seaver, G. W. Rouse, M. Obst, G. D. Edgecombe, M. V. Sørensen, S. H. D. Haddock, A. Schmidt-Rhaesa, A. Okusu, R. M. Kristensen, W. C. Wheeler, M. Q. Martindale, and G. Giribet (2008). "Broad phylogenomic sampling improves resolution of the Animal Tree of Life". In: *Nature* 452, pp. 745–749. doi: 10.1038/nature06614.
- Smith, S. A. and C. W. Dunn (2008). "Phyutility: a phyloinformatics tool for trees, alignments, and molecular data". In: *Bioinformatics* 24, pp. 715–716. doi: 10.1093/bioinformatics/btm619.
- Giribet, G., C. W. Dunn, G. D. Edgecombe, and G. W. Rouse (2007). "A modern look at the Animal Tree of Life". In: *Zootaxa* 1668, pp. 61–79.
- Oota, H., C. W. Dunn, W. C. Speed, A. J. Pakstis, M. A. Palmatier, J. R. Kidd, and K. K. Kidd (2007). "Conservative evolution in duplicated genes of the primate Class I ADH cluster". In: *Gene* 392, pp. 64–76. doi: 10.1016/j.gene.2006.11.008.
- Dunn, C. W. and G. P. Wagner (2006). "The evolution of colony-level development in the Siphonophora (Cnidaria:Hydrozoa)". In: *Development, Genes, and Evolution* 216, pp. 743–754. doi: 10.1007/s00427-006-0101-8.
- Matus, D. Q., R. R. Copley, C. W. Dunn, A. Hejnol, H. Eccleston, K. M. Halanych, M. Q. Martindale, and M. J. Telford (2006). "Broad Taxon and Gene Sampling Indicate that Chaetognaths Are Protostomes". In: *Current Biology* 16, R575–R576. doi: 10.1016/j.cub.2006.07.017.
- Matus, D. Q., K. Pang, H. Marlow, C. W. Dunn, G. H. Thomsen, and M. Q. Martindale (2006). "Deep evolutionary roots for bilaterality in the metazoa". In: *Proceedings of the National Academy of Sciences USA* 103, pp. 11195–11200. doi: 10.1073/pnas.0601257103.
- Dunn, C. W. (2005). "The complex colony-level organization of the deep-sea siphonophore *Bargmannia elongata* (Cnidaria, Hydrozoa) is directionally asymmetric and arises by the subdivision of pro-buds". In: *Developmental Dynamics* 234, pp. 835–845. doi: 10.1002/dvdy.20483.
- Dunn, C. W., P. R. Pugh, and S. H. D. Haddock (2005a). "Marrus claudanielis, a new species of deep-sea physonect siphonophore (Siphonophora, Physonectae)". In: *Bulletin of Marine Science* 76, pp. 699–714.
- Dunn, C. W., P. R. Pugh, and S. H. D. Haddock (2005b). "Molecular phylogenetics of the Siphonophora (Cnidaria), with implications for the evolution of functional specialization". In: *Systematic Biology* 54, pp. 916–935. doi: 10.1080/10635150500354837.
- Haddock, S. H. D., C. W. Dunn, and P. R. Pugh (2005). "A reexamination of siphonophore terminology and morphology, applied to the description of two new prayine species with remarkable bio-optical properties". In: *Journal of the Marine Biological Association of the U.K.* 85, pp. 695–707. doi: 10.1017/S0025315405011616.
- Haddock, S. H. D., C. W. Dunn, P. R. Pugh, and C. E. Schnitzler (2005). "Bioluminescent and red-fluorescent lures in a deep-sea siphonophore". In: *Science* 309, p. 263. doi: 10.1126/science.1110441.
- Lynch, V. J., J. J. Roth, K. Takahashi, C. W. Dunn, D. F. Nonaka, G. Stopper, and G. P. Wagner (2004). "Adaptive evolution of HoxA-11 and HoxA-13 at the origin of the uterus in mammals". In: *Proceedings of the Royal Society B: Biological Sciences* 271, pp. 2201–2207. doi: 10.1098/rspb.2004.2848.

#### d. Non-refereed journal articles

- Lewis, Z. R. and C. W. Dunn (2018). "Genome evolution: We are not so special". In: *eLife*. Not peer reviewed. doi: 10.7554/eLife.38726.
- Dunn, C. W. (2017). "Ctenophore trees". In: *Nature Ecology and Evolution*. Not peer reviewed. doi: 10.1038/s41559-017-0359-4.
- Hejnol, A. and C. W. Dunn (2016). "Animal Evolution: Are Phyla Real?" In: *Current Biology* 26. Not peer reviewed, R424–R426. doi: 10.1016/j.cub.2016.03.058.
- Dunn, C. W. (2014a). "A Cloak of Near Invisibility in an Underwater World". In: *New York Times*. Not peer reviewed. URL: <http://www.nytimes.com/2014/04/24/science/a-cloak-of-near-invisibility-in-an-underwater-world.html>.
- Dunn, C. W. (2014b). "A Marine Magician's Vanishing Act". In: *New York Times*. Not peer reviewed. URL: <http://www.nytimes.com/2014/05/28/science/a-marine-magicians-vanishing-act.html>.

- Dunn, C. W. (2014c). “Fiery Bodies Under the Waves”. In: *New York Times*. Not peer reviewed. URL: <http://www.nytimes.com/2014/08/14/science/a-colonizing-fire-of-giant-plankton.html>.
- Dunn, C. W. (2014d). “Keeping Mates Close and Competition Out in an Ocean Sponge”. In: *New York Times*. Not peer reviewed. URL: <http://www.nytimes.com/2014/11/13/science/keeping-mates-close-and-competition-out-in-an-ocean-sponge.html>.
- Dunn, C. W. (2014e). “Moving, Without Feet to Do So”. In: *New York Times*. Not peer reviewed. URL: <http://www.nytimes.com/2014/01/23/science/earth/moving-without-feet-to-do-so.html>.
- Dunn, C. W. (2014f). “Poisonous Prey Turned Into Hunter’s Defense”. In: *New York Times*. Not peer reviewed. URL: <http://www.nytimes.com/2014/02/13/science/poisonous-prey-turned-into-hunters-defense.html>.
- Dunn, C. W. (2014h). “Two Urchins, Similar but Not”. In: *New York Times*. Not peer reviewed. URL: <http://www.nytimes.com/2014/02/26/science/two-urchins-similar-but-not.html>.
- Dunn, C. W. (2013a). “As ‘Normal’ as Rabbits’ Weights and Dragons’ Wings”. In: *New York Times*. Not peer reviewed. URL: <http://www.nytimes.com/2013/09/24/science/as-normal-as-rabbits-weights-and-dragons-wings.html>.
- Dunn, C. W. (2013b). “Evolution: Out of the ocean”. In: *Current Biology* 23. Not peer reviewed, R242–R243. doi: 10.1016/j.cub.2013.01.067.
- Dunn, C. W. (2013c). “Sex in Spoonworms”. In: *New York Times*. Not peer reviewed. URL: <http://www.nytimes.com/2013/09/17/science/creatures-strange-and-complex-in-colorful-detail.html>.
- Dunn, C. W. (2013d). “The Color of Royalty, Bestowed by Science and Snails”. In: *New York Times*. Not peer reviewed. URL: <http://www.nytimes.com/2013/10/09/science/the-color-of-royalty-bestowed-by-science-and-snails.html>.
- Dunn, C. W. (2009). “Siphonophores”. In: *Current Biology* 19. Not peer reviewed, R233–R234. doi: 10.1016/j.cub.2009.02.009.
- Haddock, S. H. D. and C. W. Dunn (2005). “The complex world of siphonophores”. In: *JMBA Global Marine Environment* 2005. Not peer reviewed, pp. 24–25.

### e. Selected lectures

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|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | Intelligent Systems for Molecular Biology(ISMB) Evolution and Comparative Genomics Keynote                                                                                                                                                                                                                                                         |
| 2025 | Swarthmore College, Michigan State University, Jacques Monod Conference Metazoan Origins (Roscoff, France), North Carolina State University Genetics and Genomics Academy Seminar                                                                                                                                                                  |
| 2024 | Bio Cell Atlas Meeting (MBL)                                                                                                                                                                                                                                                                                                                       |
| 2023 | Biodiversity Cell Atlas Meeting (Barcelona, Spain), International Congress on Invertebrate Morphology (Vienna, Austria)                                                                                                                                                                                                                            |
| 2022 | Moss Landing Marine Lab, Jacques Monod Conference Metazoan Origins (Roscoff, France)                                                                                                                                                                                                                                                               |
| 2019 | Keynote speaker at the Simon Fraser University Day of Omics, Systems Biology Teory Lunch invited speaker (Harvard), Graduate Student Invited Speaker (Stony Brook, Department of Ecology and Evolution), Yale University Department of Genetics                                                                                                    |
| 2018 | UC Davis Department of Evolution and Ecology, University of Guelph Department of Integrative Biology                                                                                                                                                                                                                                               |
| 2017 | Harvard Department of Organismal and Evolutionary Biology graduate student invited speaker, University of Alabama Allele Series speaker, Missouri Botanical Garden Annual Fall Symposium invited speaker, University of Texas                                                                                                                      |
| 2016 | Stanford University Abbott Lecture, Ctenopaloosa (University of Florida) Keynote Speaker, University of Maryland BISI-BEES seminar student invited speaker, Blavatnik Science Symposium (New York Academy of Sciences), Yale University Department of Ecology and Evolutionary Biology, Boston Evolutionary Genomics Supergroup meeting at Harvard |
| 2015 | Gordon Research Conference, Friday Harbor Marine Laboratory, Systematics and Biodiversity Symposium at the Norwegian Academy of Sciences and Letters, University of Rhode Island                                                                                                                                                                   |

- 2014 Universidad Austral de Chile (Valdivia, Chile), Bowdoin College (Brunswick, Maine), University of Missouri, Evolution of First Nervous Systems II meeting (Whitney Laboratory for Marine Bioscience, Florida), 13th Annual Genome Symposium (New York University), Kyushu University (Fukuoka, Japan)
- 2013 American Museum of Natural History, Global Invertebrate Genomics Alliance (GIGA) workshop at Nova University (Fort Lauderdale, FL), meeting of the Centre National de Ressources Biologiques Marines, EMBRC (Villefranche, France), Brown Department of Ecology and Evolutionary Biology Seminar, Brown Computer Science Department Industrial Partners Program Symposium, Tree of Life Symposium at the 2013 meeting of the Society of Systematic Biologists (title: “Building trees with transcriptomes and analyzing transcriptomes with trees”), Brown EPSCoR Bioinformatics Workshop
- 2012 World Economic Forum Annual Meeting (Davos, Switzerland), Smithsonian Tropical Research Institute (Panama), Sars International Centre for Marine Molecular Biology (Bergen, Norway), Clark University, University of Miami, Stony Brook University, Applied Math at Brown University, University of Florida
- 2011 National Science Board, (Washington, DC), National Science Foundation (Washington, DC), Marine Biological Laboratory (Woods Hole, MA), Boston University (Boston, MA), Marine Observatory at Villefranche-sur-Mer (France), Smithsonian National Museum of Natural History Department of Invertebrate Zoology (Washington, DC), Smithsonian National Museum of Natural History - Symposium “Next generation Sequencing: Transformative Technology for Biodiversity Science”, keynote speaker at the Brown University Communicating Your Science Workshop, Rhode Island College
- 2010 Harvard University, University of Gothenburg (Sweden), Iowa State University, Brown University Wayland Collegium, Helicos BioSciences, National Academy of Science “Kavli Frontiers of Science” meeting, PopTech, University of Texas at Austin
- 2009 Yale University, University at Buffalo, LM University Munich, meeting of the German Research Foundation Priority Program “Deep Metazoan Phylogeny” at Humboldt University Berlin, University of Connecticut, Harvard University course OEB275r: “Phylogenetics in the Era of Genomics”, University of Rhode Island, Barcelona University, Brown MCB, invited speaker at the Society for Systematic Biologists symposium “Advances in Tree Reconstruction from Complex Data Matrices” in Moscow, Idaho
- 2008 University of Rochester, Friday Harbor Marine Laboratory, Woods Hole Oceanographic Institute, New England Biolabs
- 2007 Harvard University, Roger Williams University
- 2005 University of Hawaii

### *f. Patents and inventions*

- 2016 US Patent number 9,330,295. “Spatial sequencing/gene expression camera”

## 5 Research Grants

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### *a. Pending grants*

Collaborative Research: What creates spatial structure in open-ocean zooplankton populations? Global siphonophore genomics using recent and historical specimens, NSF, \$1,414,317, Principal Investigator.

Center: STC: Strategic Elucidation of Aquatic Slime for Innovation in Materials Emergence (SEASLIME), NSF, \$2,109,375, co-PI

*b. Current grants*

YIBS Biodiversity and Ecosystems Award, 2022, \$180,000

Ginsberg Award, Peabody Museum, 2025. \$6,000

*c. Completed grants*

“Collaborative research: The effects of predator traits on the structure of oceanic food webs”, NSF AWD0001810, September 1, 2018 - August 31, 2021, \$350,533.00, Principal Investigator.

“Real Time Phylogeny and Contact Tracing to Disrupt HIV Transmission”, NIH R01AI136058 subcontract via Miriam Hospital, 2018-2023. \$214,801

“Ginsberg Award, Peabody Museum, 2018-2021”. \$18,000

Alan T. Waterman Award, NSF, May, 2011. \$500,000 (no indirect charges)

“Inferring HIV Transmission Networks from Deep Viral Phylogenies”, Center for AIDS Research, January 1, 2017 - December 31, 2017, \$40,000, Principal Investigator.

“The evolution of gene expression and functional specialization in Siphonophora (Cnidaria)”, NSF, March 2013-March 2017. \$700,000, Principal Investigator.

“Extended phylogenetic Inference of HIV Transmission Network”, Brown University BioMed Emerging Areas of New Science Deans Awards, 2014-2015. \$80,000, co-Principal Investigator with Rami Kantor.

“Making sense of the data windfall: New statistical approaches to evolutionary analyses of gene expression”, Brown University Office of the Vice President of Research, 2013-2014. \$100,000, co-Principal Investigator with Xi Luo and Zhijin Wu.

“Collaborative Research: Resolving old questions in Mollusc phylogenetics with new EST data and developing general phylogenomic tools”, NSF, submitted July 2008, February 2009- January 2013. \$450,000 (\$775,000 total), Principal Investigator.

“IGERT: Reverse Ecology: Computational Integration of Genomes, Organisms and Environments”, NSF, submitted September, 2009. Participant.

Brown/MBL Seed Fund Award (with Joel Smith). \$12,500. Funded August, 2010.

“Phylogenetic analyses of quantitative differential expression data” Brown CCMB Seed fund. \$5,000. Funded October, 2010.

“PSCIC Full Proposal: The iPlant Collaborative: A Cyberinfrastructure-Centered Community for a New Plant Biology”, NSF, subcontract via University of Arizona, July 1, 2009 – June 30, 2012. \$257,762.

“DISSERTATION RESEARCH: Phylogeny and the Evolution of Colony Organization in the Siphonophora, Cnidaria”, NSF, June 1, 2004 – December 31, 2005, \$11,985, Co-Principal Investigator.

## 6 Service

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## 6.1 To the University

- 2025 Humanities Advisory and Tenure and Appointments Committee (HTAC), Faculty Senator, Chair of FAS-SEAS Senate Science and Engineering Committee, FASTAP Review Committee
- 2024 EEB Director of Graduate Studies (DGS), Humanities Advisory and Tenure and Appointments Committee (HTAC)
- 2023 EEB Director of Graduate Studies (DGS), CEID review committee
- 2022 EEB Director of Graduate Studies (DGS), Committee on the Economic Status of the Faculty (CESOF), EEB safety officer (overseeing COVID research reactivation and approval), Yale Genetics Training Grant (TPG) Executive Committee, Undergraduate Admissions, Yale Institute for Biospheric Studies (YIBS) small grant director
- 2021 EEB Director of Graduate Studies (DGS), Committee on the Economic Status of the Faculty (CESOF), EEB safety officer (overseeing COVID research reactivation and approval), Yale Genetics Training Grant (TPG) Executive Committee, Yale Institute for Biospheric Studies (YIBS) small grant director
- 2020 Biological Sciences Advisory Committee (July 1, 2018, to June 30, 2020), Life Sciences department review at Yale NUS, faculty search committee at Yale NUS, Committee on the Economic Status of the Faculty (CESOF), EEB safety officer (overseeing COVID research reactivation and approval), Yale Institute for Biospheric Studies (YIBS) small grant director
- 2019 Biological Sciences Advisory Committee (July 1, 2018, to June 30, 2020), Multiple committees at Yale NUS
- 2018 Biological Sciences Advisory Committee (July 1, 2018, to June 30, 2020), Peabody Exhibits Committee  
The following were at Brown University:
- 2016 Served on the Digital Teaching and Learning Committee, Director of the COBRE Computational Biology Core
- 2014 Served on the Steering Committee for the Brown Genomics Core Facility, served on the Advisory Board for the Brown Center for Computation and Visualization, maintained department website for EEB, served on the the provost's committee to explore RISD/Brown collaboration, guest lecture in BIOL2010
- 2013 Served on the University Strategic Planning Committee on Online Teaching and Learning, served on the Steering Committee for the Brown Genomics Core Facility, served on the Advisory Board for the Brown Center for Computation and Visualization, assisted in the creation of a new department website for EEB, served on the the provost's committee to explore RISD/Brown collaboration
- 2012 Participated in the Strategic Planning Sessions for the Program in Biology, served on the University Strategic Planning Committee on Online Teaching and Learning, served on the Steering Committee for the Brown Genomics Core Facility, served on the Advisory Board for the Brown Center for Computation and Visualization, presented at the Academic Technology Showcase, filed New Invention Disclosures (Tech ID 2157, 2174)
- 2011 Co-directed the Center for Computational Molecular Biology "Genome Assembly Special Forces Workshop", member of the CCV Advisory Board for High Performance Computing, member of the Illumina steering committee, was a presenter to the visiting World Economic Forum panel, presented to the Medical School Advisory Council
- 2010 Directly involved in hiring a bioinformatician in BioMed, faculty representative on panel to redesign the Brown home page, played an active role (including technical support in the lab) on the selection of a next-generation sequencer for the NIH SIG award, member of the CCV Advisory Board for High Performance Computing, member of the Illumina steering committee, participated in the EEB internal departmental review. Teaching and mentorship: sophomore advisor to two undergraduate students, MCB trainer.

- 2009 Participated in the OVPR sponsored “Brown Research Data Storage Focus Group”, provided assistance with NIH SIG proposal for a next-generation DNA sequencer, participated in discussions with IBM regarding High Performance Computing acquisition, panel member at the Sheridan Center roundtable “UTRAs – Integrating Research and Teaching in the Life & Physical Sciences”, joined the CCV Advisory Board for High Performance Computing at the invitation of the Vice President for Research, played an active role in next-generation sequencing platform selection for NIH SIG Award. Teaching and mentorship: was a sophomore advisor to two students, became a graduate student trainer in MCB, organized a special campus-wide presentation by the science writer Carl Zimmer sponsored by EEB, BIBS, and ECI
- 2008 Assisted in the preparation of a Rhode Island EPSCoR proposal for next generation sequencing and downstream computational analyses; Participated in discussions with IBM to develop supercomputing resources at Brown

## 6.2 To the profession

### Editorial

- 2025 Guest editor PLoS Computational Biology
- 2015 Guest editor at Proceedings of the National Academy of Science USA (PNAS)
- 2014 Guest editor at Proceedings of the National Academy of Science USA (PNAS), co-hosted (with Chris Pires) the Society for Systematic Biologists symposium “Phylogenomics, transcriptomics, and the evolution of gene expression” at the annual Evolution meeting
- 2012-2014 Member of the editorial board of the Journal of Experimental Zoology Part B: Molecular and Developmental Evolution

### Journal Reviews

- 2025 Bioinformatics, Journal of Comparative Neurology
- 2025 Current Biology, Nature, Nature Communications, PNAS, Mitochondrial DNA: Resources, Molecular Biology and Evolution, Current Opinion in Genetics and Development
- 2024 Molecular Biology and Evolution, Current Biology, Proceedings of the Royal Academy Series B, PNAS
- 2023 Bioinformatics Advances, Molecular Biology and Evolution, Genome Research, Evolution and Development
- 2022 PNAS, Nature Ecology and Evolutionary Biology, Nature Communications, BMC Biology, Bioinformatics, Science, Genome Biology
- 2021 PeerJ, ICOES, MBE, Nature
- 2020 PNAS, Science
- 2019 Nature, Developmental Biology, Development, PNAS
- 2018 Systematic Biology, Nature, Current Biology, Nature Ecology and Evolutionary Biology, Molecular Biology and Evolution, PeerJ, Organisms Development and Evolution
- 2017 Systematic Biology, Nature, Nature Ecology and Evolution, Nature Scientific Reports, BMC Evolutionary Biology
- 2016 Systematic Biology, Nature Scientific Reports, Current Biology, Journal of Experimental Zoology Series B, BMC Evolutionary Biology, eLIFE, PNAS
- 2015 PNAS, Current Biology, Nature Communications, EvoDevo, Systematic Biology, Proceedings of the Royal Society Series B, Nature Protocols, Nature, Zootaxa
- 2014 PLoS Currents, PNAS

|      |                                                                                                                                                                                                                                                                             |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2013 | BMC Genomics, Nature, Genome Biology and Evolution, Current Biology, PLoS One, Molecular Biology and Evolution                                                                                                                                                              |
| 2012 | Nature, PNAS, Bioinformatics, Current Biology, Molecular Reproduction and Development                                                                                                                                                                                       |
| 2011 | Current Biology, Molecular Biology and Evolution, Evolutionary Ecology, Mitochondrial DNA, PLoS One                                                                                                                                                                         |
| 2010 | Proceedings of the National Academy of Sciences (USA), Current Biology, BMC Evolutionary Biology, BMC Bioinformatics, PLoS Biology, Molecular Phylogenetics and Evolution, Trends in Genetics, Biology Letters                                                              |
| 2009 | Nature, Molecular Biology and Evolution, Current Biology, Zootaxa, BMC Evolutionary Biology, Molecular Phylogenetics and Evolution                                                                                                                                          |
| 2008 | Current Biology, BMC Evolutionary Biology, Systematic Biology, Proceedings of the Royal Society B, Journal of the Marine Biological Association of the United Kingdom, Caribbean Journal of Science, Deep-Sea Research Part I, Journal of Molecular Biology, Bioinformatics |
| 2007 | Developmental Dynamics, Hydrobiologia, Integrative and Comparative Biology, JEZ Part B: Molecular and Developmental Evolution, Journal of the Marine Biological Association of the UK, Systematics and Biodiversity, Zoologica Scripta, Zootaxa                             |

#### Other Activities

|           |                                                                                                                                                                                                                                                    |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2019-2021 | Trustee of the Society of Systematic Biologists                                                                                                                                                                                                    |
| 2016-2018 | Treasurer of the Society of Systematic Biologists                                                                                                                                                                                                  |
| 2013-2016 | Member of the supervisory board of the Neptune Consortium ( <a href="http://neptune-itsn.eu/">http://neptune-itsn.eu/</a> ), a multi-institution evolutionary developmental biology and neurobiology training network funded by the European Union |
| 2012      | Served as a faculty member for the PopTech Science Fellows Program                                                                                                                                                                                 |

## 7 Academic Honors

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|           |                                                                                    |
|-----------|------------------------------------------------------------------------------------|
| 2016      | Blavatnik Award for Young Scientists Finalist                                      |
| 2012      | Teaching with Technology Award (Brown University)                                  |
| 2011      | Alan T. Waterman Award (National Science Foundation)                               |
| 2010      | Named the Manning Assistant Professor of Ecology and Evolutionary Biology          |
| 2010      | PopTech Science and Public Leadership Fellow                                       |
| 2010      | Kavli Frontiers of Science Fellow (Sponsored by U.S. National Academy of Sciences) |
| 2006      | John Spangler Nicholas Prize (Yale University)                                     |
| 2000–2003 | NSF Graduate Research Fellowship (National Science Foundation)                     |
| 2000–2002 | Sterling Fellowship (Yale University)                                              |

## 8 Teaching

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*Courses*

|             |                                                                                                                                                                                                |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 Fall   | Invertebrate Zoology Lecture (EEB2255, 13 students)                                                                                                                                            |
| 2025 Fall   | Invertebrate Zoology Lab (EEB2256L, 13 students)                                                                                                                                               |
| 2025 Spring | Advanced Topics in Ecology and Evolution (EEB501, 5 students)                                                                                                                                  |
| 2025 Spring | Responsible Conduct of Research (EEB545, 5 students)                                                                                                                                           |
| 2024 Fall   | Phylogenetic Biology (EEB 354, 15 students)                                                                                                                                                    |
| 2024 Spring | Advanced Topics in Ecology and Evolution (EEB501, 9 students)                                                                                                                                  |
| 2024 Spring | Responsible Conduct of Research (EEB545, 9 students)                                                                                                                                           |
| 2023 Fall   | Invertebrate Zoology Lecture (EEB255, 86 students)                                                                                                                                             |
| 2023 Fall   | Invertebrate Zoology Lab (EEB256L, 13 students)                                                                                                                                                |
| 2022 Fall   | Phylogenetic Biology (EEB 354, 19 students)                                                                                                                                                    |
| 2022 Spring | Advanced Topics in Ecology and Evolution (EEB501, 10 students)                                                                                                                                 |
| 2022 Spring | Responsible Conduct of Research (EEB545, 10 students)                                                                                                                                          |
| 2021 Fall   | Advanced Topics in Ecology and Evolution (EEB500, 10 students)                                                                                                                                 |
| 2021 Fall   | Invertebrate Zoology Lecture (EEB255, 16 students)                                                                                                                                             |
| 2021 Fall   | Invertebrate Zoology Lab (EEB256L, 6 students)                                                                                                                                                 |
| 2020 Fall   | Phylogenetic Biology (EEB 354, 22 students)                                                                                                                                                    |
| 2020 Fall   | Advanced Philosophy of Biology (EEB 821, 8 students)                                                                                                                                           |
| 2019 Fall   | Philosophy of Biology (EEB321, 13 students)                                                                                                                                                    |
| 2019 Fall   | Invertebrate Zoology Lecture (EEB255, 81 students)                                                                                                                                             |
| 2019 Fall   | Invertebrate Zoology Lab (EEB256L, 9 students)                                                                                                                                                 |
| 2019 Spring | Comparative Genomics (EEB723, 11 students)                                                                                                                                                     |
| 2018 Fall   | Invertebrate Zoology Lecture (EEB255, 4 students)                                                                                                                                              |
| 2018 Fall   | Invertebrate Zoology Lab (EEB256L, 3 students)                                                                                                                                                 |
| 2018 Summer | Workshop on Molecular Evolution at the Marine Biological Laboratory (Woods Hole, MA)                                                                                                           |
| 2017 Summer | Workshop on Molecular Evolution at the Marine Biological Laboratory (Woods Hole, MA)                                                                                                           |
| 2016 Fall   | Invertebrate Zoology (Biol 0410, 43 students)                                                                                                                                                  |
| 2016 Spring | Phylogenetic Biology (Biol 1425, 13 students)                                                                                                                                                  |
| 2016 Summer | Workshop on Molecular Evolution at the Marine Biological Laboratory (Woods Hole, MA)                                                                                                           |
| 2015 Fall   | Topics in Ecology and Evolutionary Biology: interacting with data (Biol 2430, 19 students).                                                                                                    |
| 2015 Fall   | Invertebrate Zoology (Biol 0410, 37 students)                                                                                                                                                  |
| 2015 Summer | Practical Computing for Biologists Workshop, Friday Harbor Labs, University of Washington                                                                                                      |
| 2015 Summer | Workshop on Molecular Evolution at the Marine Biological Laboratory (Woods Hole, MA)                                                                                                           |
| 2014 Summer | Workshop on Molecular Evolution at the Marine Biological Laboratory (Woods Hole, MA)                                                                                                           |
| 2013 Fall   | Invertebrate Zoology (Biol 0410, 40 students)                                                                                                                                                  |
| 2013 Summer | Workshop on Molecular Evolution at the Marine Biological Laboratory (Woods Hole, MA)                                                                                                           |
| 2013 Spring | Phylogenetic Biology (Biol 1425, 11 students)                                                                                                                                                  |
| 2012 Fall   | Invertebrate Zoology (Biol 0410, 37 students)                                                                                                                                                  |
| 2012 Summer | Workshop on Molecular Evolution at the Marine Biological Laboratory (Woods Hole, MA); Guest lecture in: Bioinformatics: Biodiscovery By Computer at Summer@Brown (a high school summer course) |

|             |                                                                                                                                                                                       |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2012 Spring | EEB IGERT Course in Reverse Ecology (3 students); Guest lectures in: Analysis of Development (Biol 1310), Topics in Science Communications: Science Journalism Practicum (Biol 0950B) |
| 2011 Fall   | Invertebrate Zoology (Biol 0410, 34 students)                                                                                                                                         |
| 2011 Summer | Workshop on Molecular Evolution at the Marine Biological Laboratory (Woods Hole, MA)                                                                                                  |
| 2011 Spring | Practical Computing for Biologists NESCENT Academy at North Carolina State University (co-taught with Steve Haddock)                                                                  |
| 2010 Fall   | Invertebrate Zoology (Biol 0410, 33 students)                                                                                                                                         |
| 2010 Spring | Origins of Multicellularity and the Evolution of Germ Line (Biol 1940Y-S01, 14 students), Directed Research/Independent Study (Biol 195, 1 students)                                  |
| 2009 Fall   | Invertebrate Zoology (Biol 0410, 18 students), Directed Research/Independent Study (Biol 195, 2 students)                                                                             |
| 2009 Spring | Topics in Ecology and Evolutionary Biology (Biol 0244, 16 students, included field trip to Belize), Directed Research/Independent Study (Biol 195, 1 student)                         |
| 2008 Fall   | Topics in Ecology and Evolutionary Biology (Biol 0243, 16 students), Directed Research/Independent Study (Biol 195, 2 student)                                                        |
| 2008 Spring | Directed Research/Independent Study (Biol 0195, 2 students)                                                                                                                           |
| 2007 Fall   | Directed Research/Independent Study (Biol 0195, 1 students)                                                                                                                           |

#### *Advising—Postdoctoral researchers*

|           |                           |
|-----------|---------------------------|
| 2021–2025 | Sam Church                |
| 2020–2023 | Natasha Picciani de Souza |
| 2018–2019 | Yuanning Li               |
| 2016–2020 | Zachary Lewis             |
| 2012–2016 | Felipe Zapata             |
| 2010–2011 | Stephen Smith             |
| 2009–2015 | Stefan Siebert            |

#### *Advising—Graduate students (primary advisor)*

|              |                                                            |
|--------------|------------------------------------------------------------|
| 2023–present | Dalila Destanovic                                          |
| 2021–present | Namrata Ahuja                                              |
| 2019–2024    | Lauren Mellenthin                                          |
| 2018–2024    | Jasmine Mah                                                |
| 2015–2021    | Alejandro Damian Serrano                                   |
| 2015–2016    | Cat Luria (co-advised with Jeremy Rich)                    |
| 2013–2018    | August Guang (Applied Math, co-advised with Chip Lawrence) |
| 2013–2018    | Catriona Munro                                             |
| 2008–2015    | Rebecca Helm                                               |

*Advising—Graduate student thesis committee member*

|              |                                                                 |
|--------------|-----------------------------------------------------------------|
| 2025–present | Lan Wei (Yale, EEB)                                             |
| 2025–present | Dominica Cao (Yale, MCDB)                                       |
| 2024–present | Jacob Mathai (Yale, Immunobiology)                              |
| 2023–present | Roy Zang (Yale, MCDB)                                           |
| 2023–present | Oluwatobi Oso (Yale, EEB)                                       |
| 2023–present | Haysun Choi (Yale, EEB)                                         |
| 2021–present | Delaney Farris (Yale, Genetics)                                 |
| 2019–present | Sara Siwiecki                                                   |
| 2023–2025    | Arianna Lord (Harvard, OEB)                                     |
| 2019–2025    | Pranav Kantroo (Yale, Computational Biology and Bioinformatics) |
| 2019–2024    | Liam Taylor (Yale, EEB)                                         |
| 2019–2024    | Daniel Stadtmauer (Yale, EEB)                                   |
| 2019–2021    | Cecelia Harold (Yale, Genetics)                                 |
| 2020–2023    | Cory Berger (MIT/ WHOI)                                         |
| 2019–2023    | Chantal Parker (Yale, EEB)                                      |
| 2019–2023    | Ian Gilman (Yale, EEB)                                          |
| 2018–2021    | Jack Shaw (Yale, Geology)                                       |
| 2019–2021    | Nicolas Mongiardino Koch (Yale, Geology)                        |
| 2017–2020    | Evelyn Pless (Yale, EEB)                                        |
| 2017–2021    | Sam Church (Harvard, OEB)                                       |
| 2015–2021    | Morgan Moeglein (Yale, EEB)                                     |
| 2016–2019    | Joaquin Nunez (Brown University, EEB)                           |
| 2019–2019    | Ho Kwan Tang (Yale, EEB)                                        |
| 2016–2017    | Aislinn Rowan (Brown University, MPP)                           |
| 2015–2019    | Christy Rhine (Brown University, MCB)                           |
| 2015–2019    | Robert Lamb (Brown University, EEB)                             |
| 2015–2015    | Kara Pellowe-Wagstaff (Brown University, EEB)                   |
| 2013–2014    | Warren Francis (University of California at Santa Cruz)         |
| 2012–2017    | Tara Fresques (Brown University, MCB)                           |
| 2011–2012    | Dereck Stefanik (Boston University)                             |
| 2011–2012    | Anna Ritz (Brown University, CS)                                |
| 2011–2015    | Ariel Camp (Brown University, EEB)                              |
| 2010–2012    | Arturo Ruiz Villanueva (Institute of Ecology, Xalapa, Mexico)   |
| 2010–2015    | Cat Luria (Brown University, EEB)                               |
| 2009–2014    | Adrian Reich (Brown University, MCB)                            |
| 2009–2014    | Ben Ewen-Campen (Harvard University, OEB)                       |
| 2009–2011    | John Cumbers (Brown University, MCB)                            |
| 2009–2015    | Steven Swartz (Brown University, MCB)                           |
| 2009–2013    | Henry Astley (Brown University, EEB)                            |

2008–2012     Matt Ogburn (Brown University, EEB)  
2007–2009     James Palardy (Brown University, EEB)

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